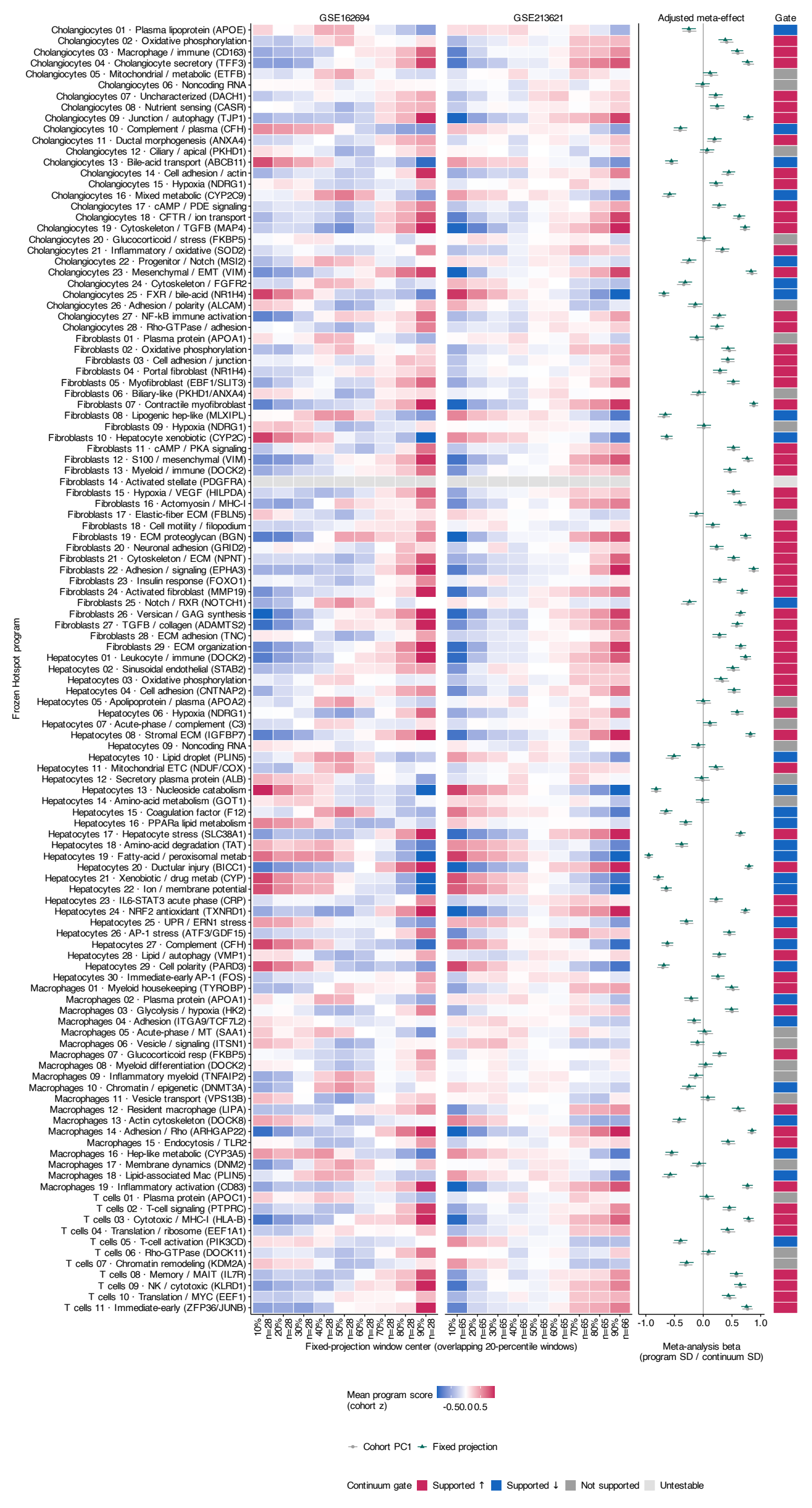
GSE162694
GSE213621
Adjusted meta-effect
Gate
Frozen Hotspot program
Cholangiocytes 01 · Plasma lipoprotein (APOE)
Cholangiocytes 02 · Oxidative phosphorylation
Cholangiocytes 03 · Macrophage / immune (CD163)
Cholangiocytes 04 · Cholangiocyte secretory (TFF3)
Cholangiocytes 05 · Mitochondrial / metabolic (ETFB)
Cholangiocytes 06 · Noncoding RNA
Cholangiocytes 07 · Uncharacterized (DACH1)
Cholangiocytes 08 · Nutrient sensing (CASR)
Cholangiocytes 09 · Junction / autophagy (TJP1)
Cholangiocytes 10 · Complement / plasma (CFH)
Cholangiocytes 11 · Ductal morphogenesis (ANXA4)
Cholangiocytes 12 · Ciliary / apical (PKHD1)
Cholangiocytes 13 · Bile-acid transport (ABCB11)
Cholangiocytes 14 · Cell adhesion / actin
Cholangiocytes 15 · Hypoxia (NDRG1)
Cholangiocytes 16 · Mixed metabolic (CYP2C9)
Cholangiocytes 17 · cAMP / PDE signaling
Cholangiocytes 18 · CFTR / ion transport
Cholangiocytes 19 · Cytoskeleton / TGFB (MAP4)
Cholangiocytes 20 · Glucocorticoid / stress (FKBP5)
Cholangiocytes 21 · Inflammatory / oxidative (SOD2)
Cholangiocytes 22 · Progenitor / Notch (MSI2)
Cholangiocytes 23 · Mesenchymal / EMT (VIM)
Cholangiocytes 24 · Cytoskeleton / FGFR2
Cholangiocytes 25 · FXR / bile-acid (NR1H4)
Cholangiocytes 26 · Adhesion / polarity (ALCAM)
Cholangiocytes 27 · NF-kB immune activation
Cholangiocytes 28 · Rho-GTPase / adhesion
Fibroblasts 01 · Plasma protein (APOA1)
Fibroblasts 02 · Oxidative phosphorylation
Fibroblasts 03 · Cell adhesion / junction
Fibroblasts 04 · Portal fibroblast (NR1H4)
Fibroblasts 05 · Myofibroblast (EBF1/SLIT3)
Fibroblasts 06 · Biliary-like (PKHD1/ANXA4)
Fibroblasts 07 · Contractile myofibroblast
Fibroblasts 08 · Lipogenic hep-like (MLXIPL)
Fibroblasts 09 · Hypoxia (NDRG1)
Fibroblasts 10 · Hepatocyte xenobiotic (CYP2C)
Fibroblasts 11 · cAMP / PKA signaling
Fibroblasts 12 · S100 / mesenchymal (VIM)
Fibroblasts 13 · Myeloid / immune (DOCK2)
Fibroblasts 14 · Activated stellate (PDGFRA)
Fibroblasts 15 · Hypoxia / VEGF (HILPDA)
Fibroblasts 16 · Actomyosin / MHC-I
Fibroblasts 17 · Elastic-fiber ECM (FBLN5)
Fibroblasts 18 · Cell motility / filopodium
Fibroblasts 19 · ECM proteoglycan (BGN)
Fibroblasts 20 · Neuronal adhesion (GRID2)
Fibroblasts 21 · Cytoskeleton / ECM (NPNT)
Fibroblasts 22 · Adhesion / signaling (EPHA3)
Fibroblasts 23 · Insulin response (FOXO1)
Fibroblasts 24 · Activated fibroblast (MMP19)
Fibroblasts 25 · Notch / RXR (NOTCH1)
Fibroblasts 26 · Versican / GAG synthesis
Fibroblasts 27 · TGFB / collagen (ADAMTS2)
Fibroblasts 28 · ECM adhesion (TNC)
Fibroblasts 29 · ECM organization
Hepatocytes 01 · Leukocyte / immune (DOCK2)
Hepatocytes 02 · Sinusoidal endothelial (STAB2)
Hepatocytes 03 · Oxidative phosphorylation
Hepatocytes 04 · Cell adhesion (CNTNAP2)
Hepatocytes 05 · Apolipoprotein / plasma (APOA2)
Hepatocytes 06 · Hypoxia (NDRG1)
Hepatocytes 07 · Acute-phase / complement (C3)
Hepatocytes 08 · Stromal ECM (IGFBP7)
Hepatocytes 09 · Noncoding RNA
Hepatocytes 10 · Lipid droplet (PLIN5)
Hepatocytes 11 · Mitochondrial ETC (NDUF/COX)
Hepatocytes 12 · Secretory plasma protein (ALB)
Hepatocytes 13 · Nucleoside catabolism
Hepatocytes 14 · Amino-acid metabolism (GOT1)
Hepatocytes 15 · Coagulation factor (F12)
Hepatocytes 16 · PPARa lipid metabolism
Hepatocytes 17 · Hepatocyte stress (SLC38A1)
Hepatocytes 18 · Amino-acid degradation (TAT)
Hepatocytes 19 · Fatty-acid / peroxisomal metab
Hepatocytes 20 · Ductular injury (BICC1)
Hepatocytes 21 · Xenobiotic / drug metab (CYP)
Hepatocytes 22 · Ion / membrane potential
Hepatocytes 23 · IL6-STAT3 acute phase (CRP)
Hepatocytes 24 · NRF2 antioxidant (TXNRD1)
Hepatocytes 25 · UPR / ERN1 stress
Hepatocytes 26 · AP-1 stress (ATF3/GDF15)
Hepatocytes 27 · Complement (CFH)
Hepatocytes 28 · Lipid / autophagy (VMP1)
Hepatocytes 29 · Cell polarity (PARD3)
Hepatocytes 30 · Immediate-early AP-1 (FOS)
Macrophages 01 · Myeloid housekeeping (TYROBP)
Macrophages 02 · Plasma protein (APOA1)
Macrophages 03 · Glycolysis / hypoxia (HK2)
Macrophages 04 · Adhesion (ITGA9/TCF7L2)
Macrophages 05 · Acute-phase / MT (SAA1)
Macrophages 06 · Vesicle / signaling (ITSN1)
Macrophages 07 · Glucocorticoid resp (FKBP5)
Macrophages 08 · Myeloid differentiation (DOCK2)
Macrophages 09 · Inflammatory myeloid (TNFAIP2)
Macrophages 10 · Chromatin / epigenetic (DNMT3A)
Macrophages 11 · Vesicle transport (VPS13B)
Macrophages 12 · Resident macrophage (LIPA)
Macrophages 13 · Actin cytoskeleton (DOCK8)
Macrophages 14 · Adhesion / Rho (ARHGAP22)
Macrophages 15 · Endocytosis / TLR2
Macrophages 16 · Hep-like metabolic (CYP3A5)
Macrophages 17 · Membrane dynamics (DNM2)
Macrophages 18 · Lipid-associated Mac (PLIN5)
Macrophages 19 · Inflammatory activation (CD83)
T cells 01 · Plasma protein (APOC1)
T cells 02 · T-cell signaling (PTPRC)
T cells 03 · Cytotoxic / MHC-I (HLA-B)
T cells 04 · Translation / ribosome (EEF1A1)
T cells 05 · T-cell activation (PIK3CD)
T cells 06 · Rho-GTPase (DOCK11)
T cells 07 · Chromatin remodeling (KDM2A)
T cells 08 · Memory / MAIT (IL7R)
T cells 09 · NK / cytotoxic (KLRD1)
T cells 10 · Translation / MYC (EEF1)
T cells 11 · Immediate-early (ZFP36/JUNB)
10% n=28
20% n=28
30% n=28
40% n=28
50% n=28
60% n=28
70% n=28
80% n=28
90% n=28
10% n=65
20% n=65
30% n=65
40% n=65
50% n=65
60% n=65
70% n=65
80% n=65
90% n=65
-1.0
-0.5
0.0
0.5
1.0
Fixed-projection window center (overlapping 20-percentile windows)
Meta-analysis beta
(program SD / continuum SD)
Mean program score
(cohort z)
-0.5 0.0 0.5
Cohort PC1
Fixed projection
Continuum gate
Supported ↑
Supported ↓
Not supported
Untestable